

ITAI 1371 Introduction to Machine Learning

Professor Viswanatha Rao (Vishwa)

Assignment: Lab 03 Exploratory Data Analysis

Group Member Contributions

Monepha Mitchell

Midterm Contributions

Ved Dalwadi

I completed the first portion of the midterm by cleaning and loading the dataset into Google Colab. I first started by importing any libraries we may need for the project then I made a raw GitHub file link to the dataset so anybody can see the code without having to download the dataset. I then printed out the dataset and looked for any major cleaning fixes. I converted Total charges from an object into numeric data. I then plotted the before and after of the Total charges and dropped the CustomerID column. Then I split it into a 70% train and 30% test split then printed new clean dataset. Finally I made the 1 page proposal for the group

Nigel Bontif,

Midterm Contributions

I contributed to the midterm project by completing the section focused on preparing the training data for machine learning. My work included cleaning repeated categorical values, such as changing “No internet service” and “No phone service” into a simpler “No” value so the data was more consistent. I also one-hot encoded the categorical columns so the model could use them as numeric inputs, and he applied StandardScaler to the numerical columns tenure, MonthlyCharges, and TotalCharges. I also created before-and-after distribution plots to show how the data looked before and after scaling. In addition, I included a Logistic Regression model using `class_weight='balanced'` to help address the class imbalance between customers who churned and customers who did not churn. Overall, my contribution helped make the training data cleaner, numeric, scaled, and more ready for future machine learning model building.

Midterm contributions

Working on this midterm project has helped me better understand the importance of data preparation and exploratory data analysis before building a machine learning model. My

role was to complete the exploratory data analysis and feature engineering on the training dataset by creating summary statistics and visualizations for contact type, tenure, payment method, and monthly charges. I also examined the class distribution and learned why class balancing is an important step before training a model. Creating new features helped me understand how additional information can improve a model's performance. Working with my group allowed us to discuss challenges and share ideas. Overall, this project increased my confidence in preparing data for machine learning and showed me how each preprocessing step contributes to building a better model.